MAT237 Notes

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Contents

1 Topology on $\mathbb{R}^n$ **3**
1.1 Interior, Boundary and Limit Points . . . 3
1.2 Sequences . . . 4
1.3 Open and Closed Sets . . . 4
1.4 Compact Sets . . . 4
1.5 Limit and Continuity . . . 5
1.6 Connected and Path-Connected Sets . . . 5

2 Differentiation **6**
2.1 Derivatives . . . 6
2.2 Gradients, Differentials and Jacobians . . . 6
2.3 The Chain Rule . . . 7
2.4 Mean Value Theorem . . . 7
2.5 Optimization . . . 7

3 Manifolds in $\mathbb{R}^n$ **8**
3.1 Tangent Space . . . 8
3.2 Diffeomorphisms . . . 8
3.3 The Inverse and Implicit Function Theorems . . . 9
3.4 Smooth Manifolds and Implicit Forms . . . 9

4 Approximation **10**
4.1 Second and Higher Order Partial Derivatives . . . 10
4.2 Taylor Polynomials . . . 11
4.3 Critical Points . . . 12

5 Integrals **12**
5.1 Partitions . . . 12
5.2 Jordan Measure . . . 13

6 Integration Methods **14**
6.1 Fubini's Theorem 14
6.2 Change of Variables 15

7 Integration on Curves **16**
7.1 Curves and Line Integrals 16
7.2 Circulation, Curl, Flux and Divergence 17
7.3 Conservative Fields and the FTLI 17
7.4 Green's Theorem 18

8 Integration on Surfaces **18**
8.1 Surfaces and Surface Integrals 18
8.2 Circulation, Curl, Flux and Divergence 19
8.3 Divergence and Classical Stokes' Theorem 20

1 Topology on $\mathbb{R}^n$

1.1 Interior, Boundary and Limit Points

Definition 1.1.1 (Interior Point)**.** Let $A \subset \mathbb{R}^n$ be a set. A point $p \in \mathbb{R}^n$ is an interior point of A if there exists $\varepsilon > 0$ such that

$$B_\varepsilon(p) \subset A.$$

Definition 1.1.2 (Interior)**.** Let $A \subset \mathbb{R}^n$ be a set. The interior of A, denoted A^o or $int(A)$, is the set of all the interior points of A.

Lemma 1.1.3. Let A and B be sets of $\mathbb{R}^n$. Then:
(a) $A^o \subseteq A$
(b) $(A \cup B)^o \supseteq A^o \cup B^o$
(c) $(A \cap B)^o = A^o \cap B^o$
(d) $(A \times B)^o = A^o \times B^o$

Definition 1.1.4 (Boundary Point)**.** Let $A \subset \mathbb{R}^n$ be a set. A point $p \in \mathbb{R}^n$ is a boundary point if for every $\varepsilon > 0$, the set $B_\varepsilon(p) \cap A \neq \emptyset$ and $B_\varepsilon(p) \cap A^c \neq \emptyset$.

Definition 1.1.5 (Topological Boundary)**.** Let $A \subset \mathbb{R}^n$ be a set. The topological boundary of A, denoted ∂A, is the set of the boundary points of A.

Lemma 1.1.6. Let A and B be sets of $\mathbb{R}^n$. Then:
(a) $\partial(A \cup B) \subseteq \partial A \cup \partial B$
(b) $\partial(A \cap B) \subseteq \partial A \cap \partial B$
(c) $\partial(A \times B) = (\partial A \times B) \cup (A \times \partial B) \cup (\partial A \times \partial B)$

Definition 1.1.7 (Limit Point)**.** Let $A \subseteq \mathbb{R}^n$ be a set. A point $p \in \mathbb{R}^n$ is a limit point of A if for every $\varepsilon > 0$, the set $B_\varepsilon(p) \setminus \{p\}$ contains points in A.

Definition 1.1.8 (Closure)**.** Let $A \subseteq \mathbb{R}^n$ be a set. The closure of A, denoted $\overline{A}$ or $cl(A)$, is the union of the set A and the set of the limit points of A.

Theorem 1.1.9. For any set $A \subseteq \mathbb{R}^n$, all of the followings hold:
(a) $A^o \subseteq A \subseteq \overline{A}$
(b) $A^o \cap \partial A = \emptyset$
(c) $\overline{A} = A^o \cup \partial A$
(d) $\partial A = \overline{A} \setminus A^o$

1.2 Sequences

Definition 1.2.1 (Sequence)**.** A sequence in $\mathbb{R}^n$ is a function with domain $\{k \in \mathbb{Z} : k \geq k_0\}$ for some fixed $k_0 \in \mathbb{Z}$ and codomain $\mathbb{R}^n$.

Definition 1.2.2 (Subsequence)**.** Let $x : \mathbb{N}^+ \to \mathbb{R}^n$ be a sequence and let $m : \mathbb{N}^+ \to \mathbb{N}^+$ be an increasing function. The sequence $\{x_{m_k}\}_{k=1}^{\infty}$ is a subsequence of the sequence $\{x_k\}_{k=1}^{\infty}$.

Definition 1.2.3 (Convergence of a Sequence)**.** Fix a point $p \in \mathbb{R}^n$. A sequence $\{x_k\}_{k=1}^{\infty}$ converges to p if:

$$\forall \varepsilon > 0, \exists K \text{ s.t. } \forall k \in \mathbb{N}, k \geq K \Rightarrow \|x_k - p\| < \varepsilon.$$

We write $\lim_{k\to\infty} x_k = p$ if this holds.

1.3 Open and Closed Sets

Definition 1.3.1 (Open Set)**.** A set $A \subseteq \mathbb{R}^n$ is open if every point in A is an interior point of A.

Definition 1.3.2 (Closed Set)**.** A set $A \subseteq \mathbb{R}^n$ is closed if every limit points of A belongs to A.

Theorem 1.3.3. A set $A \subseteq \mathbb{R}^n$ is open iff its complement $A^C = \mathbb{R}^n \backslash A$ is closed.

1.4 Compact Sets

Definition 1.4.1 (Compact Set)**.** A set $A \subseteq \mathbb{R}^n$ is compact if every sequence of A has a subsequence which convergeces to $a \in A^\circ$.

Definition 1.4.2 (Bounded and Unbounded Sets)**.** A set $A \subseteq \mathbb{R}^n$ is bounded if there exists $R > 0$ such that $A \subseteq \{x \in \mathbb{R}^n : \|x\| < R\}$. If A is not bounded, then A is unbounded.

Theorem 1.4.3 (Heine-Borel Theorem)**.** A set $A \subseteq \mathbb{R}^n$ is compact iff A is closed and bounded.

Theorem 1.4.4 (Bolzano-Weierstrass Theorem)**.** Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

1.5 Limit and Continuity

Definition 1.5.1 (Limit)**.** Let $A \subseteq \mathbb{R}^n$ and $f : A \to \mathbb{R}^m$. Let a be an interior point of A. Let $b \in \mathbb{R}^m$, then b is the limit of f at a if:

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } 0 < \|x - a\| < \delta \Rightarrow \|f(x) - b\| < \varepsilon$$

Definition 1.5.2 (Continuity)**.** Let $A \subseteq \mathbb{R}^n$ and $f : A \to \mathbb{R}^m$. Let a be an interior point of A, then f is continuous at a if:

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } 0 < \|x - a\| < \delta \Rightarrow \|f(x) - f(a)\| < \varepsilon$$

Theorem 1.5.3 (Squeeze Theorem)**.** Let $A \subseteq \mathbb{R}^n$ be a set and let a be a limit point of A. Let f, g, h be real-valued functions with domain A. Assume that there exists $\delta > 0$ such that:

$$\forall x \in A, \|x - a\| < \delta \Rightarrow f(x) \leq g(x) \leq h(x).$$

If $\lim_{x \to a} f(x) = \lim_{x \to a} h(x) = k$, then $\lim_{x \to a} g(x) = k$.

1.6 Connected and Path-Connected Sets

Definition 1.6.1 (Path-Connected Set)**.** A set $S \subseteq \mathbb{R}^n$ is path-connected if for all $p, q \in S$, there exists curve $\gamma : [a, b] \to S$ such that $\gamma(a) = p, \gamma(b) = q$ and $\text{im}(\gamma) \subseteq S$.

Definition 1.6.2 (Convex Set)**.** A set $S \subseteq \mathbb{R}^n$ is convex if for all $t \in [0, 1]$ and $p, q \in S$, $tp + (1 - t)q \in S$.

Theorem 1.6.3. Let $S \subseteq \mathbb{R}^n$ be a path-connected set. Let f be a $\mathbb{R}^m$-valued function defined on S. If f is continuous on S, then f is path-connected on S.

Theorem 1.6.4 (Intermediate Value Theorem)**.** Let f be a real-valued function defined on $[a, b]$. If f is continuous on $[a, b]$, then $f([a, b])$ is path-connected.

Theorem 1.6.5 (Extreme Value Theorem)**.** Let $A \subset \mathbb{R}^n$ and $f : A \to \mathbb{R}$. If A is non-empty and compact and f is continuous on A, then f attains its maxima and minima on A.

Theorem 1.6.6 (Least Upper Bound Principle)**.** Let $S \subseteq \mathbb{R}$ be a non-empty subset of real numbers. If S has a upper bound, then the suprema of S exists.

2 Differentiation

2.1 Derivatives

Definition 2.1.1 (j^{th} Partial Derivative)**.** Let $A \subseteq \mathbb{R}^n$ and $f : A \to \mathbb{R}^m$. Let a be an interior point of A. Let $\{e_1, \cdots, e_n\}$ be the standard basis vectors for $\mathbb{R}^n$. Then, the jth partial derivative of f at a is given by:

$$\partial_j f(a) = \lim_{h \to 0} \frac{f(a + he_j) - f(a)}{h}$$

Definition 2.1.2 (Directional Derivative)**.** Let $A \subseteq \mathbb{R}^n$ and $f : A \to \mathbb{R}^m$. Let a be an interior point of A. Let $u \in \mathbb{R}^n$ be a vector, then the directional derivative at a is given by:

$$D_u f(a) = \lim_{h \to 0} \frac{f(a + hu) - f(a)}{h}$$

2.2 Gradients, Differentials and Jacobians

Definition 2.2.1 (Gradient)**.** Let $A \subseteq \mathbb{R}^n$ and $f : A \to \mathbb{R}$. Let a be an interior point of A, then the gradient of f at a is given by:

$$\nabla f(a) = (\partial_1 f(a), \cdots, \partial_n f(a))$$

Definition 2.2.2 (Differential)**.** Let $A \subseteq \mathbb{R}^n$ and $f : A \to \mathbb{R}^m$. Let a be an interior point of A. We say f is differentiable at a if there exists a linear map $M : \mathbb{R}^n \to \mathbb{R}^m$ such that:

$$\lim_{h \to 0} \frac{f(a + h) - f(a) - Mh}{\|h\|} = 0,$$

and the linear map M is called the differential of f at a, denoted by df_a.

Theorem 2.2.3. Let $A \subseteq \mathbb{R}^n$ and let $f : A \to \mathbb{R}^m$ be a function. Let a be an interior point of A. If f is differentiable at a, then for all $u \in \mathbb{R}^n$, the directional derivative $D_u f(a)$ exists, and:

$$df_a(u) = D_u f(a)$$

Definition 2.2.4 (Jacobian)**.** Let $A \subset \mathbb{R}^n$ and let $f : A \to \mathbb{R}^m$ be a function. Let a be an interior point of A. Let $f = (f_1, \cdots, f_m)$. The Jacobian of f at a is defined by:

$$Df(a) = [\partial_i f_i(a)]_{ij} = \begin{pmatrix} \partial_1 f_1(a) & \cdots & \partial_n f_1(a) \\ \vdots & \ddots & \vdots \\ \partial_1 f_m(a) & \cdots & \partial_n f_m(a) \end{pmatrix} = \begin{pmatrix} \nabla f_1(a) \\ \vdots \\ \nabla f_m(a) \end{pmatrix}$$

Theorem 2.2.5. Let $A \subseteq \mathbb{R}^n$ and $f : A \to \mathbb{R}^m$ be a function. Let a be an interior point of A. Let $u \in \mathbb{R}^n$, then:

$$Df_a(u) = Df(a) \cdot u$$

Definition 2.2.6 (C^1)**.** Let $A \subseteq \mathbb{R}^n$ and $f : A \to \mathbb{R}^m$. Let a be an interior point of A. Then, f is continuously differentiable at a (or C^1 at a) if $\partial_1 f, \cdots, \partial_n f$ are all defined in a neighborhood of a and are all continuous at a.

2.3 The Chain Rule

Theorem 2.3.1 (The Chain Rule)**.** Let $A \subseteq \mathbb{R}^n$ and $B \subseteq \mathbb{R}^m$. Let a be an interior point of A. If $f : A \to B$ and $g : B \to \mathbb{R}^k$ are differentiable at a and $f(a)$ respectively, then $h = g \circ f$ is differentiable at a, and:

$$dh_a = dg_{f(a)} \circ df_a,$$

and the Jacobians satisfy:

$$Dh(a) = Dg(f(a)) \cdot Df(a)$$

Corollary 2.3.2. Let $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ be open sets. If the function $f : U \to V$ and $g : V \to \mathbb{R}^k$ are both C^1, then the function $g \circ f : U \to \mathbb{R}^k$ is C^1.

2.4 Mean Value Theorem

Theorem 2.4.1 (Mean Value Theorem)**.** Let $U \subseteq \mathbb{R}^n$ be open and let $a, b \in U$. Let f be a differentiable real-valued function on U. If U contains the line segment L from a to b, then there exists $c \in L$ such that:

$$\nabla f(c) \cdot (b - a) = f(b) - f(a)$$

2.5 Optimization

Theorem 2.5.1 (Local Extreme Value Theorem)**.** Let $A \subseteq \mathbb{R}^n$ and let $f : A \to \mathbb{R}^m$. Let a be an interior point of A. If f has a local extrema at a, then either $\nabla f(a) = 0$ or $\nabla f(a)$ does not exist.

Definition 2.5.2 (Critical Point)**.** Let $A \subseteq \mathbb{R}^n$ and let $f : A \to \mathbb{R}^m$. Let a be an interior point of A, then a is a critical point of f if either $\nabla f(a) = 0$ or $\nabla f(a)$ does not exist.

Definition 2.5.3 (Saddle Point)**.** Let $A \subseteq \mathbb{R}^n$ and let $f : A \to \mathbb{R}^m$. Let a be an interior point of A, then a is a saddle point of f if $\nabla f(a) = 0$ and f does not have an extrema at a.

Theorem 2.5.4 (Lagrange Multiplier Theorem)**.** Let U be an open subset of $\mathbb{R}^n$. Let $g : U \to \mathbb{R}$ and $f : U \to \mathbb{R}$ be C^1 real-valued functions. Fix $c \in \mathbb{R}$, defined:

$$S = \{x \in U : g(x) = c\}.$$

Assume $\nabla g(p) \neq 0$ for any $p \in S$. If f has a local extrema on S at the point a, then there exists $\lambda \in \mathbb{R}$ such that:

$$\nabla f(a) = \lambda \nabla g(a)$$

The quantity λ is called the Lagrange multiplier.

3 Manifolds in $\mathbb{R}^n$

3.1 Tangent Space

Definition 3.1.1 (Tangent Vector)**.** Let $S \subseteq \mathbb{R}^n$ be a set and let p be a point in S. A vector $v \in \mathbb{R}^n$ is the tangent vector of S at p if there exists an open interval $I \subseteq \mathbb{R}$ containing 0 and a map $\gamma : I \to \mathbb{R}^n$ such that γ is C^1, $\gamma(I) \subseteq S$, $\gamma(0) = p$ and $\gamma'(0) = v$.

Definition 3.1.2 (Tangent Space)**.** Let $S \subseteq \mathbb{R}^n$ be a set and let p be a point in S. The tangent space of S at p, denoted T_pS, is the set of tangent vectors of S at p.

Definition 3.1.3 (Tangent Plane)**.** Let $S \subseteq \mathbb{R}^n$ be a set and let $p \in S$. The tangent plane of S at p, denoted $p + T_pS$, is the tangent space translated to p.

3.2 Diffeomorphisms

Definition 3.2.1 (Global Inverse)**.** Let U and V be subsets of $\mathbb{R}^n$. The global inverse of $F : U \to V$ is the map $g : V \to U$ satisfying $G \circ F(x) = x$ for all $x \in U$ and $F \circ G(y) = y$ for all $y \in V$, i.e.

$$\forall x \in U, \forall y \in V, y = F(x) \iff x = G(y).$$

The inverse of F is unique and denoted F^{-1}.

Definition 3.2.2 (Global Diffeomorphism)**.** Let U and V be open subset of $\mathbb{R}^n$. A function $F : U \to V$ is a global diffeomorphism if F is bijective, C^1 and its inverse function $F^{-1} : V \to U$ is C^1.

Definition 3.2.3 (Local Diffeomorphism)**.** Let A and B be open subsets of $\mathbb{R}^n$. Fix a point $a \in A$. A function $F : A \to B$ is a local diffeomorphism at a if there exists an open subset $U \subseteq A$ containing a, such that $F(U)$ is open, and the restriction $F|_U : U \to F(U)$ is a global diffeomorphism. The function $G = F|_U^{-1} : F(U) \to U$ is a local inverse of F at a.

Lemma 3.2.4. Let A and B be open subsets of $\mathbb{R}^n$. If $F : A \to B$ is a global diffeomorphism, then F is a local diffeomorphism at every point $a \in A$.

3.3 The Inverse and Implicit Function Theorems

Theorem 3.3.1 (Inverse Function Theorem, pt.1)**.** Let U and V be open subsets of $\mathbb{R}^n$. Let the function $F : U \to V$ be a diffeomorphism. Then for every $x \in U$, the Jacobian $DF(x)$ is an invertible $n \times n$ matrix and the Jacobian of the inverse function $G = F^{-1}$ satisfies:

$$DG(y) = D^{-1}F(x) \ \forall x \in U \text{ and } y = F(x)$$

Theorem 3.3.2 (Inverse Function Theorem, pt.2)**.** Let A and B be open subsets of $\mathbb{R}^n$. Fix $a \in A$. Let $F : A \to B$ be a C^1 function. If the Jacobian $DF(a)$ is invertible, then F is a local diffeomorphism at a.

Theorem 3.3.3 (Implicit Function Theorem)**.** Let $U \subseteq \mathbb{R}^n \times \mathbb{R}$ be open and let f be a real-valued C^1 function on U. Let $a \in \mathbb{R}^n, b \in R$ and $(a, b) \in U$. If $f(a, b) = 0$ and $\dfrac{\partial f}{\partial y}(a, b) \neq 0$, then the equation $f(x, y) = 0$ defines y locally as a C^1 function ϕ of x near (a, b).

Theorem 3.3.4 (Implicit Function Theorem for Several Variables)**.** Let $U \subseteq \mathbb{R}^n \times \mathbb{R}^k$ be open. Let $F : U \to \mathbb{R}^k$ be a C^1 map. Let $a \in \mathbb{R}^n, b \in \mathbb{R}^k$ so $(a, b) \in U$. If $F(a, b) = 0$ and the $k \times k$ matrix:

$$\frac{\partial F}{\partial y}(a, b) = \frac{\partial(F_1, ..., F_k)}{\partial(y_1, ..., y_k)}(a, b) = \left[\frac{\partial F_i}{\partial y_j}(a, b)\right]_{ij}$$

is invertible, then the equation $F(x, y) = 0$ locally defines $y = (y_1, ..., y_k)$ as a $\mathbb{R}^k$-valued C^1 function $x = (x_1, ..., x_n)$ near (a, b).

3.4 Smooth Manifolds and Implicit Forms

Definition 3.4.1 (k-Dimensional Smooth Manifold)**.** Fix $k, n \in \mathbb{N}^+$ with $k < n$. Let $S \subseteq \mathbb{R}^n$ be a set and let point $p \in S$. The set S is a k-dimensional smooth manifold at p if:

(1) There exists an open set $U \subseteq \mathbb{R}^n$ such that $p \in U$.

(2) $S \cap U$ is the graph of a C^1 function $f : V \to \mathbb{R}^{n-k}$, where $V \subseteq \mathbb{R}^k$ is open.

Moreover, S is a k-dimensional smooth manifold if S is a k-dimensional smooth manifold at every point $p \in U$.

Theorem 3.4.2. Let S and S' be subsets of $\mathbb{R}^n$ with $p \in S \cap S'$. If there exists an open set $U \subseteq \mathbb{R}^n$ containing p such that $S \cap U = S' \cap U$, then $T_pS = T_pS'$.

Theorem 3.4.3. Let $U \subseteq \mathbb{R}^n$ be an open set.Let $f : U \to \mathbb{R}$ be a C^1 function. Assume the set:

$$S = \{x \in U : f(x) = 0\} = f^{-1}(0),$$

is non-empty. Fix $p \in S$, if $\nabla f(p) \neq 0$, then S is a $(n-1)$-dimensional smooth manifold at p iff $\nabla f(p) \cdot v = 0$. That is,

$$T_pS = \{v \in \mathbb{R}^n : \nabla f(p) \cdot v = 0\},$$

therefore, the tangent plane of S of p is given by:

$$p + T_pS = \{x \in \mathbb{R}^n : \nabla f(p) \cdot (x - p) = 0\}.$$

Definition 3.4.4 (Implicit Form)**.** Let $m, n \in \mathbb{N}^+$ with $n > m$. A set $S \subseteq \mathbb{R}^n$ can be written in implicit form if there exists a constant $c \in \mathbb{R}^m$, a set $A \subseteq \mathbb{R}^n$, and a continuous function $f : A \to \mathbb{R}^m$ such that:

$$S = f^{-1}(\{c\}) = \{x \in \mathbb{R}^n : f(x) = c\}$$

4 Approximation

4.1 Second and Higher Order Partial Derivatives

Definition 4.1.1 (Second Order Partial Derivative)**.** Let $U \subseteq \mathbb{R}^n$ be open. Let $f : U \to \mathbb{R}^m$ be C^1. Fix $i, j \in \{1, ..., n\}$, then the second order partial derivative $\partial i \partial j f$ at a is defined by:

$$\partial i \partial j f(a) = \partial i(\partial j f)(a)$$

Provided that the ith partial of $\partial j f$ exists at a.

Definition 4.1.2 (Twice Continuously Differentiable)**.** Let $U \subseteq \mathbb{R}^n$ be an open set. Let $f : U \to \mathbb{R}^m$ be a function, then f is twice continuously differentiable or C^2 if $\partial i \partial j f$ exists and is continuous everwhere in U.

Theorem 4.1.3 (Clairaut's Theorem)**.** Let $U \subseteq \mathbb{R}^n$ and $f : U \to \mathbb{R}^m$ be C^2. Then $\partial i \partial j f = \partial j \partial i f$ for all $i, j \in \{1, ..., n\}$.

Definition 4.1.4 (Hessian Matrix)**.** Let f be a real valued function which is C^2 at $a \in \mathbb{R}^n$. The Hessian of f at a is the $n \times n$ matrix $Hf(a) = [\partial i \partial j f(a)]_{ij}$.

Definition 4.1.5 (k-Times Continuously Differentiable)**.** Let $k \in \mathbb{N}^+$. Let $U \subseteq \mathbb{R}^n$ be an open set. Let $f : U \to \mathbb{R}^m$ be k-times continuously differentiable or C^k, then for all $i_1, i_2, ..., i_k \in \{1, ..., n\}$, its kth order partial derivatives $\partial_{i_1} \partial_{i_2} ... \partial i_k f$ exist and continuous everywhere in U.

Lemma 4.1.6. Every polynomial is C^∞.

Theorem 4.1.7 (Generalized Clairaut's Theorem)**.** Let $U \subseteq \mathbb{R}^n$ be open. Let $f : U \to \mathbb{R}^m$ be C^k, then the mixed partial derivatives of f up to order k commute. That is, for any integer $1 \leq i_1, i_2, ..., i_k \leq n$ and any reordering $j_1, j_2, ..., j_k$ of $i_1, i_2, ..., i_k$, there is:

$$\partial i_1 \partial i_2 ... \partial i_k f = \partial j_1 \partial j_2 ... \partial j_k f.$$

Definition 4.1.8 (Multi-Index)**.** An element $\alpha = (\alpha_1, ..., \alpha_n) \in \mathbb{N}^n$ is a multi-index. The degree of α is a non-negative integer $|\alpha| = \alpha_1 + ... + \alpha_n$. The factorial $\alpha! = \alpha_1! \cdot ... \cdot \alpha_n!$.

Definition 4.1.9 (α-Partial Derivative)**.** Let $U \subseteq \mathbb{R}^n$ be an open set. Let $f : U \to \mathbb{R}^m$ be C^k. For any multi-index $\alpha = (\alpha_1, ..., \alpha_n) \in \mathbb{N}^n$ with degree $|\alpha| \leq k$, define the α-partial derivative by $\partial^\alpha f = \partial_1^{\alpha_1} f \cdot ... \cdot \partial_n^{\alpha_n} f$.

Definition 4.1.10 (Monomial)**.** Let $\alpha = (\alpha_1, ..., \alpha_n) \in \mathbb{N}^+$ be a multi-index, and let $x = (x_1, ..., x_n) \in \mathbb{R}^n$. The monomial x^α in the variables $x_1, ..., x_n$ is defined by $x^\alpha = x_1^\alpha \cdot ... \cdot x_n^\alpha$, where x^α has degree $|\alpha| = \alpha_1 + ... + \alpha_n$ in the variables $x_1, ..., x_n$.

Definition 4.1.11 (Polynomial)**.** A polynomial is a finite linear combination of monomials. The degree of a polynomial is the maximum degree of its monomials.

4.2 Taylor Polynomials

Definition 4.2.1 (k^{th}-Taylor Polynomial)**.** Let f be a function of class C^k on an open ball centered at $a \in \mathbb{R}^n$. The kth Taylor Polynomial of f at a is:

$$P_k(x) = \sum_{\substack{a \in \mathbb{N}^n \\ |a| \leq k}} \frac{\partial^k f(a)}{k!}(x - a)^k$$

Definition 4.2.2 (k^{th} Order Approximation)**.** Let f and g be real-valued functions defined on an open ball centered at $a \in \mathbb{R}^n$. The function g is an kth order approximation of f at $a \in \mathbb{R}^n$ if:

$$\lim_{x \to a} \frac{f(x) - g(x)}{\|x - a\|^k} = 0$$

Theorem 4.2.3 (Taylor's Theorem)**.** Let f be a real-valued function of class C^k on an open ball at $a \in \mathbb{R}^n$. A polynomial P is the kth Taylor Polynomial of f at a if and only if P is the unique degree $\leq k$ polynomial which is the kth order approximation of f at a.

4.3 Critical Points

Definition 4.3.1 (Quadratic Form)**.** Let A be a real $n \times n$ symmetric matrix. The quadratic form associated to A is the function $q : \mathbb{R}^n \to \mathbb{R}$ given by $q(n) = v^T Av$ for $v \in \mathbb{R}^n$.

Lemma 4.3.2. Let A be a real $n \times n$ symmetric matrix. Let $q : \mathbb{R}^n \to \mathbb{R}$ be its associated quadratic form. If $v \in \mathbb{R}^n$ is an eigenvector of A with eigenvalue $\lambda \in \mathbb{R}$, then $q(v) = \lambda \|v\|^2$

Theorem 4.3.3 (Spectral Theorem)**.** Every $n \times n$ real symmetric matrix has an orthonormal basis of eigenvectors with real eigenvalues.

Theorem 4.3.4. Let A be a $n \times n$ real symmetric matrix.Let $q : \mathbb{R}^n \to \mathbb{R}$ be its corresponding quadratic form, where $q(v) = v^T Av$ for $v \in \mathbb{R}^n$. Then, the maximum and minimum values of q on the unit sphere S^{n-1} are respectively equal to the maximum and minimum eigenvalues of A.

Theorem 4.3.5 (Second Derivative Test)**.** Let f be a real-value function of class C^2 on an open ball at $a \in \mathbb{R}^n$. If a is a critical point of f, the f has:

(a) a local minimum if all the eigenvalues of the Hessian $Hf(a)$ are positive.
(b) a local maximum if all the eigenvalues of the Hessian $Hf(a)$ are negative.
(c) a saddle point if the Hessian $Hf(a)$ has positive and negative eigenvalues.

5 Integrals

5.1 Partitions

Definition 5.1.1. A rectangle in $\mathbb{R}$ is a closed interval $[a, b]$, where $a, b \in \mathbb{R}$ and $a < b$. The length of $[a, b]$ is defined to be $b - a$.

Definition 5.1.2 (Partition)**.** A partition P of the interval $[a, b]$ is a finite set such that $\{a, b\} \subseteq P \subseteq [a, b]$.

Definition 5.1.3. Write $P = \{x_0, ..., x_k\}$, where $a = x_0 < ... < x_k = b$. Then:

(a) The index set of P is the set $\{1, ..., k\}$.
(b) For $i \in \{1, ..., k\}$, the i-subrectangle of P is given by $R_i = [x_{i-1}, x_i]$.
(c) The subrectangles of P is the set of rectangles $\{R_1, ..., R_k\}$.

Definition 5.1.4. A rectangle in $\mathbb{R}^2$ is a closed interval $R = [a, b] \times [c, d]$, where $a, b, c, d \in \mathbb{R}$ and $a < b, c < d$. The area of R is defined to be $(b - a)(d - c)$.

Definition 5.1.5. A rectangle in $\mathbb{R}^n$ is a closed interval $R = [a_1, b_1] \times ... \times [a_n, b_n]$, where $a1,..., a_n, b_1, ..., b_n \in \mathbb{R}$ and $a_i < b_i \forall 1 \leq i \leq n$. The volume of R is defined to be $(b_1 - a_1) \cdot ... \cdot (b_n - a_n)$.

Theorem 5.1.6. Let R be a rectangle in $\mathbb{R}^n$. Let P be a partition of R. Let I be the index set of P and let $\{R_i : i \in I\}$ be the subrectangles of P. Then:

(a) $R = \bigcup_{i \in I} R_i$

(b) $\forall i, j \in I, i \neq j \Rightarrow \text{int}(R_i \cap R_j) = \emptyset$

(c) $\text{vol}(R) = \sum_{i \in I} \text{vol}(R_i)$

Definition 5.1.7 (Refinement)**.** Let $P = (P_1, \cdots, P_n)$ and $P' = (P'_1, \cdots, P'_n)$ be partitions of a rectangle R. The partition P' is a refinement of P if $P_j \subseteq P'_j$ for every $j \in \{1, \cdots, n\}$.

5.2 Jordan Measure

Definition 5.2.1 (Indicator Function)**.** Let $S \subseteq \mathbb{R}^n$. The indicator function of S is the function $\chi_S : \mathbb{R}^n \to \mathbb{R}$ defined by:

$$\chi_S(x) = \begin{cases} 1, x \in S \\ 0, x \notin S \end{cases}$$

Definition 5.2.2 (Jordan Measure)**.** A set $S \subseteq \mathbb{R}^n$ is Jordan measurable if there exists a rectangle $R \subseteq \mathbb{R}^n$ and χ_S is integrable on R. In this case, the Jordan measure of S is:

$$\text{vol}(S) = \int_R \chi_S dV$$

Theorem 5.2.3. Let $S \subseteq \mathbb{R}^n$ and let $R, R' \subseteq \mathbb{R}^n$ be rectangles containing S. Then χ_S is integrable on R if and only if χ_S is integrable on R'.
In this case:

$$\int_R \chi_S dV = \int_{R'} \chi_S dV$$

Theorem 5.2.4. A set $S \subseteq \mathbb{R}^n$ is Jordan Measurable if and only if S is bounded, ∂S is Jordan measurable and $\text{vol}(\partial S) = 0$.

Theorem 5.2.5. Let $S, T, S_1, ..., S_k \subseteq \mathbb{R}^n$ be Jordan measurable sets. Then:

(a) If $S \subseteq T$, then $\text{vol}(S) \leq \text{vol}(T)$.

(b) The sets $S \cap T$ and $S \cup T$ are Jordan measurable, and:

$$\text{vol}(S \cup T) = \text{vol}(S) + \text{vol}(T) - \text{vol}(S \cap T).$$

(c) The union $S_1 \cup \cdots \cup S_k$ is Jordan measurable, and:

$$\text{vol}(S_1 \cup \cdots \cup S_k) \leq \text{vol}(S_1) + \text{vol}(S_2) + \cdots + \text{vol}(S_k).$$

Theorem 5.2.6. Let $S \subseteq \mathbb{R}^n$ be Jordan measurable. Then $\bar{S}, S^o$ and ∂S are Jordan measurable and $\text{vol}(S) = \text{vol}(\bar{S}) = \text{vol}(S^o)$, and $\text{vol}(\partial S) = 0$.

Definition 5.2.7 (Jordan Measure Zero)**.** A set $S \subseteq \mathbb{R}^n$ is Jordan measure zero if $\text{vol}(S) = 0$.

Theorem 5.2.8. A set $S \subseteq \mathbb{R}^n$ is Jordan measure zero if and only if for every $\varepsilon > 0$, there exists rectangles $R_1, \cdots, R_N \subseteq \mathbb{R}^n$ such that:

$$S \subseteq \bigcup_{i=1}^{N} R_i, \ \sum_{i=1}^{N} \text{vol}(R_i) < \varepsilon$$

Theorem 5.2.9. Any finite subset $S \subseteq \mathbb{R}^n$ is Jordan measure zero.

Lemma 5.2.10. Let $S \subseteq \mathbb{R}^n$. If $S^0 \neq \emptyset$, then S does not have Jordan measure zero.

Lemma 5.2.11. Let $S, S_1, \cdots, S_k \subseteq \mathbb{R}^n$ be Jordan measure zero sets. Then:

(a) Every subset of S has Jordan measure zero.

(b) The union $S_1 \cup \cdots \cup S_k$ has Jordan measure zero.

(c) The closure $\bar{S}$ has Jordan measure zero.

Theorem 5.2.12 (Sard's Theorem)**.** Let $k, n \in \mathbb{R}^+$ with $k < n$. Let $R \subseteq \mathbb{R}^k$ be a rectangle and let $f : R \to \mathbb{R}^n$ be C^1 on an open set containing R. Then the image $f(R) = \{f(x) : x \in R\}$ is Jordan measurable with $\text{vol}(f(R)) = 0$.

6 Integration Methods

6.1 Fubini's Theorem

Lemma 6.1.1. Let $f : [a, b] \to \mathbb{R}$ be a bounded real-valued function on the interval $[a, b]$. If f is integrable on $[a, b]$, then:

$$\int_{[a,b]} f \mathrm{d}V = \int_a^b f(x) \mathrm{d}x.$$

Definition 6.1.2 (Slices)**.** Let $f : [a, b] \times [c, d] \to \mathbb{R}$ be a function. A x-slice of f at $x = x_0$ is a function:

$$g : [c, d] \to \mathbb{R} \text{ s.t. } g(y) = f(x_0, y)$$

for some fixed $x_0 \in [a, b]$. Similarly, a y-slice of f at $y = y_0$ is a function:

$$h : [a, b] \to \mathbb{R} \text{ s.t. } h(x) = f(x, y_0)$$

for some fixed $y_0 \in [c, d]$.

Lemma 6.1.3. Let $f : [a,b] \times [c,d] \to \mathbb{R}$ be a function. If f is bounded or continuous, then every slice of f is bounded or continuous.

Theorem 6.1.4 (Fubini's Theorem)**.** Let $f : [a,b] \times [c,d] \to \mathbb{R}$ be bounded.

(a) For $x \in [a,b]$, define $f^x : [c,d] \to \mathbb{R}$ by $f^x(y) = (x,y)$. Assume f^x is integrable on $[c,d]$ for every $x \in [a,b]$ and f is integrable on $[a,b] \times [c,d]$. Then $\displaystyle\int_c^d f(x,y)\mathrm{d}y$ and $\displaystyle\int_a^b \left(\int_c^d f(x,y)\mathrm{d}y\right)\mathrm{d}x$ exists for every $x \in [a,b]$, and:
$$\iint_R f\mathrm{d}A = \int_a^b \left(\int_c^d f(x,y)\mathrm{d}y\right)\mathrm{d}x.$$

(b) If f is continuous, then $\displaystyle\int_a^b \left(\int_c^d f(x,y)\mathrm{d}y\right)\mathrm{d}x$ and $\displaystyle\int_c^d \left(\int_a^b f(x,y)\mathrm{d}x\right)\mathrm{d}y$ both exist, and:
$$\iint_{[a,b]\times[c,d]} f\mathrm{d}A = \int_a^b \left(\int_c^d f(x,y)\mathrm{d}y\right)\mathrm{d}x = \int_c^d \left(\int_a^b f(x,y)\mathrm{d}x\right)\mathrm{d}y.$$

Theorem 6.1.5. Let $\Omega \subseteq \mathbb{R}^2$ be a Jordan measurable set such that the restricted polar coordinate transformation $g|_\Omega : \Omega \to g(\Omega)$ is a bijection. If $f : g(\Omega) \to \mathbb{R}$ is integrable on $g(\Omega)$, then $F : \Omega \to \mathbb{R}$ given by $F(r,\theta) = (f \circ g)(r,\theta) \cdot |r|$ is integrable on Ω, and:
$$\iint_{g(\Omega)} f\mathrm{d}A = \iint_\Omega F\mathrm{d}A.$$

Theorem 6.1.6 (Fubini's Theorem in 3D)**.** Let $R = [a,b] \times [c,d] \times [e,f]$ and let $\varphi : R \to \mathbb{R}$ be bounded. If the x-slice of φ is integrable on $[c,d] \times [e,f]$, the xy-slice of φ is integrable on $[e,f]$, and φ is integrable on R, then every iterated triple integral exists and is equal to $\displaystyle\iiint_R \varphi dV$.

Theorem 6.1.7 (n-Dimensional Fubini's Theorem)**.** Let $R = [a_1,b_1] \times \cdots \times [a_n,b_n]$ be a rectangle in $\mathbb{R}^n$. Let $f : R \to \mathbb{R}$ be a bounded real-value function. If f is integrable on R and every slice of f on R is integrable on its domain, then every iterated n-fold integral exists and are all equal to $\displaystyle\int_R f dV$.

6.2 Change of Variables

Theorem 6.2.1. Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a linear map. For every rectangle R in $\mathbb{R}^n$:
$$\mathrm{vol}(T(R)) = |\det(T)|\,\mathrm{vol}(R)$$

Theorem 6.2.2 (Change of Variable Theorem)**.** Let $U, V \subseteq \mathbb{R}^n$ be open sets. Let $g : U \to V$ be a diffeomorphism. Let $\Omega \subseteq U$ be a Jordan measurable set. Assume $|\det(Dg)|$ is bounded on Ω. The function f is integrable on $g(\Omega)$ iff $(f \circ g)|\det(Dg)|$ is integrable on Ω. If so:

$$\int_{g(\Omega)} f dV = \int_{\Omega} (f \circ g)|\det(Dg)| dV$$

7 Integration on Curves

7.1 Curves and Line Integrals

Definition 7.1.1 (Smooth Curve)**.** A curve $\gamma : [a, b] \to \mathbb{R}^n$ is smooth if γ is C^1 on $[a, b]$ and γ' is bounded on $[a, b]$.

Definition 7.1.2 (Regular Curve)**.** A curve $\gamma : [a, b] \to \mathbb{R}^n$ is regular if γ is differentiable on (a, b) and $\gamma'(t) \neq 0$ for all $t \in [a, b]$.

Definition 7.1.3 (Simple Curve)**.** A curve $\gamma : [a, b] \to \mathbb{R}^n$ is simple if for all $s, t \in (a, b)$, $s \neq t \Rightarrow \gamma(s) \neq \gamma(t)$.

Definition 7.1.4 (Closed Curve)**.** A curve $\gamma : [a, b] \to \mathbb{R}^n$ is closed if $\gamma(a) = \gamma(b)$.

Theorem 7.1.5. If $\gamma : [a, b] \to \mathbb{R}^n$ is a smooth, regular and simple parametrization of a set $C \subseteq \mathbb{R}^n$, then the set C is a 1-dimensional smooth manifold at $p = \gamma(c)$ for every $c \in (a, b)$. Moreover, the tangent space satisfies $T_pC = \text{span}\{\gamma'(c)\}$.

Definition 7.1.6 (Arc Length)**.** The arc length of a curve $C \subseteq \mathbb{R}^n$ is defined as:

$$l(C) = \int_a^b \|\gamma'(t)\| dt,$$

where C is parametrized by $\gamma : [a.b] \to \mathbb{R}^n$.

Definition 7.1.7 (Line Integral)**.** Let C be a curve in $\mathbb{R}^n$ parametrized by $\gamma : [a, b] \to \mathbb{R}^n$. Let f be a bounded real-valued function defined on C. The line integral of f over C is defined by:

$$\int_C f ds = \int_a^b f(\gamma(t))\|\gamma'(t)\| dt$$

If the line integral of f over C exists, then f is integrable on C.

7.2 Circulation, Curl, Flux and Divergence

Definition 7.2.1 (Vector Field)**.** A n-dimensional vector field is a function $F : \mathbb{R}^n \to \mathbb{R}^n$.

Definition 7.2.2 (Divergence in $\mathbb{R}^2$)**.** Let $F = (F_1, F_2)$ be a C^1 vector field in $\mathbb{R}^2$ and let $p = (x, y)$ be a point in $\mathbb{R}^2$, then the divergence of F at p equals to:

$$(\nabla \cdot F)(p) = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y}$$

Definition 7.2.3 (Curl in $\mathbb{R}^2$)**.** Let $F = (F_1, F_2)$ be a C^1 vector field in $\mathbb{R}^2$, and let $p = (x, y)$ be a point in $\mathbb{R}^2$, then the curl of F at p is:

$$(\nabla \times F)(p) = \begin{vmatrix} \partial_x & \partial_y \\ F_1 & F_2 \end{vmatrix} = \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}$$

Definition 7.2.4 (Flux in $\mathbb{R}^2$)**.** Let F be a C^1 vector field in $\mathbb{R}^2$ defined on an orientated closed curve C in $\mathbb{R}^2$. Then the flux across C is defined as $\oint_C F \cdot n ds$.

Definition 7.2.5 (Circulation in $\mathbb{R}^2$)**.** Let F be a C^1 vector field in $\mathbb{R}^2$ defined on an orientated closed curve C in $\mathbb{R}^2$. Then the circulation across C is defined as $\oint_C F \cdot T ds$.

Definition 7.2.6 (Unit Tangent and Unit Normal)**.** Let C be an orientated curve in $\mathbb{R}^2$, parametrized by $\gamma(t) = (x(t), y(t))$ for $\gamma : [a, b] \to \mathbb{R}^2$ and $a \leq t \leq b$, then:

$$T(t) = \frac{(x'(t), y'(t))}{\|\gamma'(t)\|} \text{ and } n(t) = \frac{(y'(t), -x'(t))}{\|\gamma'(t)\|}$$

7.3 Conservative Fields and the FTLI

Definition 7.3.1 (Conservative Field)**.** A vector field F is conservative on an open set $U \subseteq \mathbb{R}^n$ if there exists a function f such that $\nabla f = F$ on U. f is called the potential function of F.

Definition 7.3.2 (Irrotational Vector Field)**.** A C^1 vector field F is irrotational on an open set $U \subseteq \mathbb{R}^n$ if $(\nabla \times F)(p) = \vec{0}$ for all $p \in U$.

Lemma 7.3.3. Let F be a C^1 vector field on $U \subseteq \mathbb{R}^n$. If U is convex and F is irrotational on U, then F is conservative on U.

Theorem 7.3.4 (The Fundamental Theorem of Line Integrals)**.** Let C be an orientated curve in $\mathbb{R}^n$ parametrized by $\gamma : [a, b] \to \mathbb{R}^n$. Let f be a real-valued function that is C^1 on an open set containing C, then:

$$\int_C \nabla f \cdot d\gamma = f(\gamma(b)) - f(\gamma(a))$$

Definition 7.3.5 (Simply Connected Set)**.** A set $D \subseteq \mathbb{R}^2$ is simply connected if:

(a) D is path-connected.

(b) For every closed curves $C \subseteq D$, the region enclosed by C is contained in D.

Theorem 7.3.6. Let $F : U \subseteq \mathbb{R}^2 \to \mathbb{R}^2$ be a C^1 vector field, where U is open. If U is simply connected and F is irrotational on U, then F is conservative on U.

7.4 Green's Theorem

Theorem 7.4.1 (Green's Theorem, Curl Form)**.** Let F be a C^1 vector field on a regular region $R \subseteq \mathbb{R}^2$. If the boundary ∂R is a finite disjoint union of positively orientated piecewise curves, then:

$$\oint_{\partial R} F \cdot T ds = \iint_R (\nabla \times F) dA$$

Equivalently, let $F = (P, Q)$, so we can also express the theorem as:

$$\oint_{\partial R} P dx + Q dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Theorem 7.4.2 (Green's Theorem, Flux Form)**.** Let F be a C^1 vector field on a regular region $R \subseteq \mathbb{R}^2$. If the boundary ∂R is a finite disjoint union of positively orientated piecewise curves, then:

$$\oint_{\partial R} F \cdot n ds = \iint_R (\nabla \cdot F) dA$$

8 Integration on Surfaces

8.1 Surfaces and Surface Integrals

Definition 8.1.1 (Surface Area)**.** Let $S \subseteq \mathbb{R}^3$ be a surface parametrized by $G : U \to \mathbb{R}^3$. The surface area of S is defined as:

$$A(S) = \iint_U \|\partial_1 G \times \partial_2 G\| dA$$

Theorem 8.1.2 (Invariance Of Surface Area)**.** Let $G : U \to S$ and $H : V \to S$ be parametrizations of the surface $S \subseteq \mathbb{R}^3$. If G is a parametrization of H, then:

$$\iint_U \|\partial_1 G \times \partial_2 G\| dA = \iint_V \|\partial_1 H \times \partial_2 H\| dA$$

Definition 8.1.3 (Surface Area)**.** Let $S \subseteq \mathbb{R}^3$ be a surface parametrized by $G : U \to \mathbb{R}^3$, then the surface area equals to:

$$A(S) = \iint_S 1 dS,$$

where $dS = \|\partial_1 G \times \partial_2 G\| dA$.

Definition 8.1.4 (Scalar Surface Integral)**.** Let $S \subseteq \mathbb{R}^3$ be a surface parametrized by $G : U \to \mathbb{R}^3$. Let f be a bounded real-valued function on S. The scalar surface integral of f over S is given by:

$$\iint_S f dS = \iint_U (f \circ G) \|\partial_1 G \times \partial_2 G\| dA$$

Definition 8.1.5 (Orientated Surface)**.** An orientated surface S is a set of parametrizations that are reparametrizations of each other with the same orientations.

Definition 8.1.6 (Unit Normal)**.** The unit normal n of an orientated surface $S \subseteq \mathbb{R}^3$ is a countinuous function $n : S \to \{w \in \mathbb{R}^3, \|w\| = 1\}$, which for any parametrization $G : U \to \mathbb{R}^3$ of the orientated surface S, is defined by:

$$n(G(u, v)) = \frac{(\partial_1 G \times \partial_2 G)(u, v)}{\|(\partial_1 G \times \partial_2 G)(u, v)\|}$$

8.2 Circulation, Curl, Flux and Divergence

Definition 8.2.1 (Divergence in $\mathbb{R}^3$)**.** Let $F = (F_1, F_2, F_3)$ be a C^1 vector field in $\mathbb{R}^3$, then the divergence of F at point $p = (x, y, z)$ is:

$$(\nabla \cdot F)(p) = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

Definition 8.2.2 (Sourceless Vector Field)**.** Let F be a C^1 vector field in $\mathbb{R}^3$. A point $p \in \mathbb{R}^3$ is a source if $(\nabla \cdot F)(p) > 0$. A vector field F is sourceless if for all $p \in \mathbb{R}^3$, $(\nabla \cdot F)(p) = 0$.

Definition 8.2.3 (Curl in $\mathbb{R}^3$)**.** Let F be a C^1 vector field in $\mathbb{R}^3$, then the curl of F at point $p = (x, y, z)$ is:

$$(\nabla \times F)(p) = \begin{vmatrix} e_1 & e_2 & e_3 \\ \partial_x & \partial_y & \partial_z \\ F_1 & F_2 & F_3 \end{vmatrix} = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial x}, \frac{\partial F_3}{\partial x} - \frac{\partial F_1}{\partial z}, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right)$$

Definition 8.2.4 (Flux in $\mathbb{R}^3$)**.** Let F be a C^1 vector field on a regular region $S \subseteq \mathbb{R}^3$. Then the flux over F on S is defined as $\iint_S F \cdot n dS$.

Definition 8.2.5 (Circulation in $\mathbb{R}^3$)**.** Let F be a C^1 vector field on a orientated closed curve $C \subseteq \mathbb{R}^3$. Then the circulation over F on C is defined as $\oint_C F \cdot T ds$.

8.3 Divergence and Classical Stokes' Theorem

Theorem 8.3.1 (Divergence Theorem)**.** Let F be a C^1 vector field on a regular region $V \subseteq \mathbb{R}^3$. If its boundary ∂V is a finite disjoint union of closed piecewise curves, then:

$$\oiint_{\partial V} F \cdot n dS = \iiint_V (\nabla \cdot F) dV$$

Theorem 8.3.2 (Classical Stokes' Theorem)**.** Let $S \subseteq \mathbb{R}^3$ be a surface orientated with unit normal n and whose boundary ∂S is a finite disjoint union of closed piecewise curves. Let F be a C^1 vector field in $\mathbb{R}^3$ on an open set containing S. Then:

$$\oint_{\partial S} F \cdot T ds = \iint_S (\nabla \times F) \cdot n dS.$$